UNRWA - Education Department (TVET)
Gaza Training Centre
Blended Learning Course Outline
Commercial Departments

Course Description

Title	Mobile Applications Development Using Flutter	Code	SICT235
Department / Level	Programming and Database Level 2 (PDB2)	Semester	Third Semester (Winter 2026-2027)
Prerequisites	OOP; Web Development and Design	Instructor	Eng. Abeer Elaydi
Duration	16 weeks	Allocation	3 sessions/week; 2 hours/session
Practical / Theory	2:1	Total	42 teaching sessions

Course Overview

Dart and Flutter development covering widget composition, styling, responsive layouts, lists and grids, interaction, navigation, validated forms, Provider, local preferences, asynchronous programming, JSON and REST APIs. The course introduces Streams and concludes with project refinement, documentation and application delivery.

Entry requirement: prior programming and OOP knowledge; planned access to a development PC or laptop.

Course Objectives

1. Configure Flutter, run a project and apply typed Dart, null safety, functions, collections, OOP and basic asynchronous programming.
2. Construct reusable, styled and responsive interfaces with assets, lists, grids and appropriate layout constraints.
3. Implement interactions, local state, navigation, data passing and validated forms.
4. Manage shared state with Provider and save, retrieve and restore simple preferences using SharedPreferencesAsync.
5. Inspect JSON; implement REST CRUD requests and integrate loading, error, empty and success states.
6. Distinguish Future from Stream and adapt a guided StreamBuilder example.
7. Use documentation to debug code; explain independent work, prepare a README and verify an APK; recognize introductory PWA and notification workflows.

Methods and Tools

- Google Classroom, Telegram and Google Meet; remote and in-person practical sessions.
- VS Code, Flutter/Dart SDKs, DartPad, Android tools and Postman.

Training Strategies

- Guided blended learning.
- Live coding.
- Independent practice.
- Progressive project-based learning.
- Debugging.
- Documentation tasks.
- Code review.
- Individual feedback.

Course Outline

14 teaching weeks / 42 sessions; final assessment in Weeks 15-16

D = Dart; F = Flutter; T = tools; P = package documentation. Reference links appear on page 3.

Week	Topics	Practical Competency	References
01	**Dart foundations and setup in parallel** Dart types, null safety, control flow, List/Map; setup in parallel.	Write and debug typed Dart code; verify setup.	D1; F1
02	**Dart for widgets and documentation literacy** Functions, callbacks, Dart OOP; first run; introductory Future and async/await; documentation.	Create objects, run an app and trace a simple Future.	D1-D2; F2; F13; F15-F18
03	**Widget fundamentals, styling and layout** Widget trees, MaterialApp/Scaffold, theme, assets and Row/Column.	Build a styled screen from a design reference.	F2; F10; F13-F14
04	**Composition, constraints and flexible sizing** Reusable widgets, constraints, Expanded/Flexible and overflow repair.	Extract components and explain layout repairs.	F3; F11; F13
05	**Responsive screens, lists and grids** Responsive sizing, bounded scrolling, ListView.builder and GridView.builder.	Check lists/grids at different sizes and text scales.	F3-F4; F13
06	**Interaction, local state and item identity** StatefulWidget, callbacks, setState; guided Keys and ValueKey.	Implement local updates; apply a guided stable key.	F2; F7; F19-F21
07	**Navigation and multi-screen integration** Navigator.push/pop, route data and independent multi-screen practice.	Connect screens, pass data and repair navigation.	F5
08	**Midterm practical and targeted feedback**		
09	**User input and validated forms** Text input, controller lifecycle, Form, form keys and validation.	Reject invalid input and process valid submissions.	F6
10	**Provider and shared application state** State ownership, ChangeNotifier, Provider and shared collections.	Keep shared add/remove actions consistent across screens.	F7
11	**Asynchronous Dart and local preferences** Async consolidation; SharedPreferencesAsync, defaults and restoration.	Save/read a preference and restore it after restart.	D2; F7; F23
12	**JSON, REST requests and Postman** JSON models; GET, POST, PUT, PATCH and DELETE; headers/body, status codes and async errors.	Inspect and implement basic REST requests with JSON and error handling.	D2-D3; F8; T1
13	**FutureBuilder, API integration and Streams introduction** FutureBuilder states/retry; CRUD integration and refresh after changes; guided Streams/StreamBuilder.	Handle responses, refresh data after changes and adapt a stream example.	D4; F7-F9; F22
14	**Delivery and platform features** APK, web/PWA and local notifications; parallel project practice.	Verify an APK and adapt guided platform examples.	F11-F12; F24-F25; P1
15-16	**Final Assessment**	Final practical, project demonstration and individual discussion.	

Assessment and References

Evaluation and Grading Policy

Assessment Component	Marks
Assignments & Labs	20
Quizzes, Meeting Attendance	10
Interviews & Orals	10
Project	10
Mid-Exam	20
Final Exam	30
Total	100

Important to Consider

- Attendance for all class meetings is required. Students are responsible for checking topics and assignment information if they miss a class.
- Submissions and executions will not be accepted after the activity has been graded and returned to the class or once the solution has been posted.
- Submitted work must include appropriate attribution and demonstrate individual understanding and independent work.

Learning References

Flutter and Dart Fundamentals. Shereef Abu Al-Maati. Apress, 2026.

Flutter for Beginners, Fourth Edition. Thomas Bailey & Alessandro Biessek. Packt, 2026.

Flutter Cookbook, Third Edition. Packt, 2026. Selected Dart and Flutter recipes.

Flutter Apprentice, Fourth Edition. Kodeco, 2024. Supplementary reading.

Technical References

Selected sections and examples; linked titles below. F = official Flutter resources, including Flutter-maintained packages; D = Dart; T = Postman; P = package publisher.

D1 | Language
D2 | Future, async and await
D3 | JSON support
D4 | Streams tutorial
F1 | Installation
F2 | Widgets
F3 | Constraints
F4 | Responsive design
F5 | Navigation
F6 | Forms
F7 | State management
F8 | Networking
F9 | FutureBuilder
F10 | Assets and images
F11 | Debugging
F12 | Android release
F13 | API reference
F15 | Official videos
F16 | Flutter Samples
F17 | Material examples
F18 | Learning resources
F19 | Key
F20 | ValueKey
F21 | Keyed lists example
F22 | StreamBuilder example
F23 | SharedPreferencesAsync examples
F24 | Web build
F25 | Web / PWA FAQ
T1 | Postman quick start
P1 | Local notifications